Syed Fuzail Farhan

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Education

University of Waterloo | BASc, Mechatronics Engineering (Co-op) 2023 – 2028

Technical Skills

Firmware: C, C++, Python; STM32 HAL/CubeIDE, ESP32, micro-ROS, FreeRTOS; ADC, PWM, GPIO interrupts
Interfaces and tools: UART/USART, Ethernet/UDP, LwIP, Git, Linux, oscilloscopes, multimeters
Hardware: DRV8353 and DRV8833 motor drivers, servo and stepper motors

Engineering Experience

Undergraduate Research Assistant | University of Waterloo Oct 2026 – Present
- Beginning research under **Prof. Sanjeev Bedi** on **two TETRIX robots** navigating independently on a grid; project goal is to stop on mutual detection and communicate to resolve potential path conflicts.
- Planned sensing: **4–6 cameras per robot** for **360-degree coverage**, using a **multiplexer** for a shared output.

Firmware Developer | UW Robotics Team (Mars Rover) Jun 2026 – Present
- Established **micro-ROS message exchange** between an **STM32 NUCLEO-F439ZI** and a host Agent over **USB/USART3**, using FreeRTOS firmware and DMA-backed serial transport.
- Configured **LwIP/UDP, RMII Ethernet, and static IP addressing** for migration from serial transport; verified board reachability from both **Windows and Ubuntu/WSL** before testing Agent communication.
- Investigating missing Agent traffic with **STM32CubeIDE breakpoints, DMA status registers, and tcpdump** to isolate issues in Ethernet receive-buffer handling, firmware transmission, and host-side packet delivery.
- Developing the STM32 interface for a rover with **two NVIDIA Jetsons**, aiming to move **time-sensitive I/O and preprocessing** off the Ubuntu-based arm and rover computers.

Research Assistant | IDEAS Clinic, University of Waterloo May 2026 – Aug 2026
- Enabled vision-guided part sorting for the Toyota Innovation Challenge by integrating a **Dobot arm**, **Orbbec depth camera**, and **Python/OpenCV** colour-segmentation pipeline.
- Prototyped a **YOLO hand-detection** condition that pauses arm motion while a hand is detected in the work area; tested changes in object colour and pick/drop locations.
- Supported **short R&D turnaround times** by rapidly prototyping, testing, and troubleshooting engineering demonstrations to meet **project deadlines**.
- Reduced contact-stress error against AGMA calculations from **76.2% to 5.13%** by investigating mesh convergence and rebuilding an **ANSYS Mechanical** contact model for a spur-gear study.
- Built **Python ingestion pipelines** for UW DataHub, combining environmental readings from **REST APIs**, web exports, and emailed reports for campus researchers.

Research Assistant | IDEAS Clinic, University of Waterloo Aug 2025 – Dec 2025
- Developed a **four-day microcontroller/FPGA workshop** for **60+ students**, covering PWM, serial communication, digital I/O, and circuit debugging with oscilloscopes and multimeters.
- Co-designed a **three-day ANSYS FEA/CFD workshop** delivered to **70+ students** using engineering case studies and physical test models.
- Designed and **3D-printed a heat-exchanger test assembly**; measured **airflow and temperatures** to set simulation inputs and investigate discrepancies in a co-authored Ansys education case study.

Software Engineering Intern | GrazeAI May 2024 – Aug 2024
- Extracted data from **50+ mobile conference apps in roughly 30 minutes** by building automation pipelines and generalizing extraction by app developer for reuse across apps from the same provider.
- Reduced manual trial identification from **days to minutes** with a reusable **Python/API workflow** that filtered **600,000+ clinical trials** against custom criteria.

Selected Project

Two-Axis Motor Control | STM32F401RE, HAL, Embedded C Completed lab
- Enabled potentiometer-based **stepper-motor speed control** on a two-axis positioning apparatus by extending **STM32 HAL firmware** to read and map ADC input.
- Implemented **polling and interrupt-driven GPIO synchronization** and **limit-switch handling** to stop motion at the ends of travel.